

Tutorial 1 Project Management Concept

Question 1

Differentiate between a software project and a construction project by completing Table 1 below. Include an example in each of your answers.

Table 1: Comparison of software project and construction project

	Software Project (example: developing the Exam Timetable System)	Construction Project (example: building a bridge)
Invisibility	Progress is not immediately visible because the project has no physical form. Example: A project manager may find it difficult to show the "physical" progress of the Exam Timetable System's backend logic to a client.	Progress is immediately visible because the project has a tangible physical form. Example: The progress of a bridge is clear to everyone as the pillars and structure are physically erected over time
Flexibility	Software is generally easier to change throughout its development. Example: If requirements change, it is relatively easy to update the Exam Timetable System to include a new room-assignment rule	It is difficult to change a construction project once it has been completed. Example: Once a bridge is built, it is nearly impossible to change its foundation or fundamental structural design

Question 2

T-Shirts2U is a small company that supplies and manufactures customized t-shirts. However, its current system, which is partially automated, is unable to keep track of the orders and stocks efficiently. Besides that, it is unable to provide accurate information to the management on which items are more profitable, and there have been mistakes in some of the order fulfillments. In order to solve these problems, the senior management of T-Shirts2U has requested you to develop a new online ordering system to increase the company's annual sales by RM50,000. The new system to be developed has the following modules: Customer, Order carts, Online payment, and Order fulfillment. This project must be completed within 3 months with an allocated budget of RM180,000. The project team consists of six software developers.

a) Identify and explain TWO (2) constraints of the T-shirt2U project implementation. Explain your answer.

1. Time Constraints

- This project have the strict deadline 3 months and must be completed within 3 months
- The project only have 3 months to complete the customer, order carts, online payment and order fulfillment modules
- delay may increase sales and postpone the expected sales growth
- proper scheduling and resources allocation are required to ensure it can be completed on time

2. Budget Constraints

- This project have the allocated fixed budget of RM180,000
- All development activities, testing process, maintenance fee cannot exceed and completed within this budget

-Project manager need to keep track of the use of budget to avoid overspending

-any additional feature or unexpected problems might require extra resources, which may lead to exceed the budget

b) The outcome of a project can be classified as truly successful, partially successful, or a total failure. Hence, the new online ordering system project can be considered as a total failure if the Customer, Order carts, Online payment, and Order fulfillment modules not completed within 3 months, the costs exceed the fixed budgets of RM180,000 and fail to hit the target of increase the company's annual sales by RM50,000.

Question 3

Return on Investment (ROI), Net Present Value (NPV) and many other financial techniques can be used to evaluate and compare the risks and rewards of different projects. They help project managers to decide whether a project is worthwhile to pursue or not. Table 1 illustrates the cash flow of Project ABC. The cash flow takes place at the end of each year.

Table 1: Cash flow of Project ABC

Year	Project X	Discount Factor	Discounted cash flow or Present Value
0	-50,000	$1/(1+0.12)^0 = 1.0000$	-50,000
1	20,000	$1/(1+0.12)^1 = 0.8929$	$20000 * 0.8929 = 17858$
2	20,000	$1/(1+0.12)^2 = 0.7972$	$20000 * 0.7972 = 15944$
3	20,000	$1/(1+0.12)^3 = 0.7118$	$20000 * 0.7118 = 14236$
4	30,000	$1/(1+0.12)^4 = 0.6355$	$30000 * 0.6355 = 19065$
NPV			17103

NPV: $17858 + 15944 + 14236 + 19065 - 50000 = 17103$

Using a 12% discount rate, calculate the Net Present Value (NPV) and Return on Investment (ROI) for this Project. Based on your calculations, analyse whether the project is attractive or should be rejected. Explain your answer.

Note:

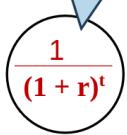
- Show your workings in the calculation.
- Negative values represent costs.
- Discount factors round up to 4 decimal points. All other figures round up to the nearest integer.
- ROI in percentage, rounds up to 2 decimal points.

$$\text{ROI} = \frac{\text{average annual profit}}{\text{total investment}} \times 100$$

$$\text{ROI} = \text{NPV} / \text{Total Discounted Cost} \times 100\%$$

How to calculate the Present Value of a future cash flow ?

Present Value

of a future cash flow = Cash Flow in Year t X 

where

- **r** is the **discount rate** (expressed as a decimal value)
- **t** is the **number of years into the future** that the cash flow occurs

$$NPV = RM\ 17103$$

$$ROI = (17103/50000) * 100\% = 0.34206 = 34.21\%$$

Question 4

Using a 10% discount rate, calculate the Net Present Value (NPV) and Return on Investment (ROI) for Project X and Project Y. Based on your calculations, suggest the most profitable project. Explain your answer.

Show your calculation to justify your answer. Discount factors round up to 4 decimal points.

Discount Rate: 10% for both projects					
Project X	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	50,000	40,000	30,000	-
Discount Factor	1	$1/(1+0.10)^1 = 0.9090$	$1/(1+0.10)^2 = 0.8265$	$1/(1+0.10)^3 = 0.7513$	
Discounted Costs	100,000	$50000 * 0.9090 = 45450$	$40000 * 0.8265 = 33060$	$30000 * 0.7513 = 22539$	$100000 + 45450 + 33060 + 22539 = 201049$
Benefits	0	100,000	120,000	100,000	-
Discount Factor	1	$1/(1+0.10)^1 = 0.9090$	$1/(1+0.10)^2 = 0.8265$	$1/(1+0.10)^3 = 0.7513$	
Discounted Benefits	0	90900	99180	75130	$90900 + 99180 + 75130 = 265210$
NPV	$265210 - 201049 = 64161$				
ROI: Use discounted value	$ROI = NPV / \text{Total Discounted Cost} \times 100\%$ $64161 / 201049 * 100\% = 0.3191 = 31.91\%$				

Project Y	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	70,000	60,000	20,000	-
Discount Factor	1	$1/(1+0.10)^1 = 0.9090$	$1/(1+0.10)^2 = 0.8265$	$1/(1+0.10)^3 = 0.7513$	
Discounted Costs	100,000	63630	49590	15026	$100000 + 63630 + 49590 + 15026 = 228246$

Benefits	0	100,000	150,000	90,000	-
Discount Factor	1	$1/(1+0.10)^1 = 0.9090$	$1/(1+0.10)^2 = 0.8265$	$1/(1+0.10)^3 = 0.7513$	
Discounted Benefits	0	90900	123975	67617	$90900 + 123975 + 67617 = 282492$
NPV	$282492 - 228246 = 54246$				
ROI:	$\text{ROI} = \text{NPV} / \text{Total Discounted Cost} \times 100\%$ $54246 / 228246 \times 100\% = 0.2377 = 23.77\%$				

Comment:

Question 5

ABC Tech Company provides software consultancy services to business clients. It focuses on selling accounting software. The company consists of 4 different departments: **sales and marketing**, **customer services**, **information technology (IT)**, and **finance**. Each staff member will be assigned to a department according to his or her area of specialization and expertise. The **sales and marketing** department consists of 3 staff, 2 staff work in the **customer service department**, while 3 staff work in the **finance department**, and 3 experienced staff work in the **IT department**. Each department is supervised by a functional manager, and each functional manager in turn reports to the Chief Executive Officer (CEO).

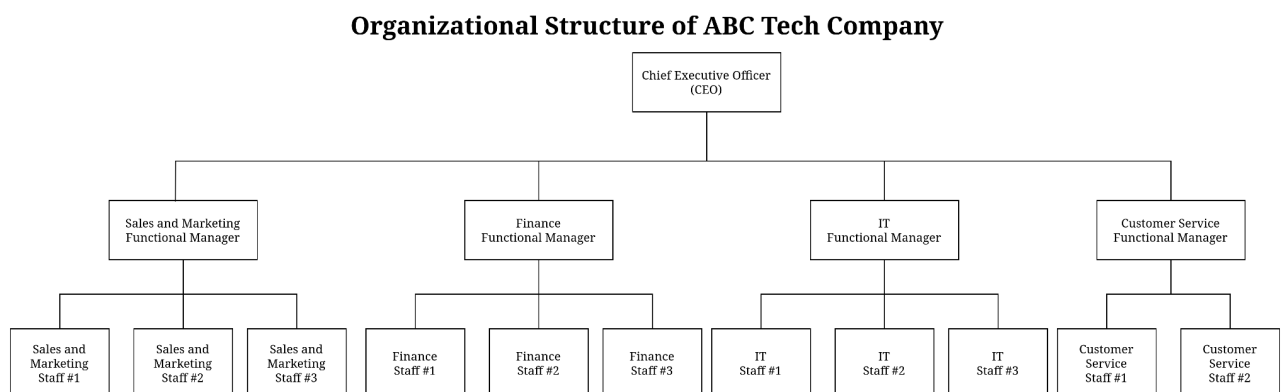
sales and marketing department : 3

customer service department : 2

finance department : 3

IT department : 3

a) Draw the organization structure of this company.



b) Based on part a), explain TWO (2) advantages of this organizational structure.

- This is *functional organizational structure*.
- 1. **Cost Savings:** by grouping resources together by function, the company can avoid the duplication of facilities and resources that might occur in different departments.
- 2. **Specialization and Expertise :** this structure allows staff to work in their area of **Specialization and Expertise**. They can improve their knowledge and skills in their specific job area, which helps them produce better quality work.

Question 6

You are a project manager of an online education management system (EMS) developed for a private college. After communicating with the staff for collecting requirements, the following scope is defined: the system will be used to manage communication between lecturers and students, uploading and downloading learning materials (such as tutorial questions, lecture notes, assignments specifications), making announcements, and assessing students' assignments. The modules of EMS can be modularized with a well-defined scope and high staff involvement. The project team focuses on the core features of the user interface design, assessment, communication, announcement, and submission. A budget of RM150,000 has been approved for the project with a duration of 8 weeks.

Construct a *Work Breakdown Structure* (WBS) for the online education management system using an outline numbering format. Assume that you have chosen the *Waterfall software process model*.

1.0 Requirements Analysis

- 1.1 Gather requirements from stakeholders
 - 1.1.1 Interview lecturers and students
 - 1.1.2 Document functional requirements
 - 1.1.3 Document non-functional requirements
- 1.2 Analyze system requirements
 - 1.2.1 Define user roles (lecturer, student, admin)
 - 1.2.2 Specify module requirements
 - 1.2.3 Define data requirements
- 1.3 Develop Software Requirements Specification (SRS)
- 1.4 Validate requirements with stakeholders

2.0 System Design

- 2.1 Design system architecture
- 2.2 Design User Interface
 - 2.2.1 Design login and registration screens
 - 2.2.2 Design dashboard interface
 - 2.2.3 Design navigation and layout
- 2.3 Design database schema
- 2.4 Design module specifications
 - 2.4.1 Design assessment module
 - 2.4.2 Design communication module
 - 2.4.3 Design announcement module
 - 2.4.4 Design submission module
 - 2.4.5 Design learning materials module
- 2.5 Review and approve design

3.0 Implementation

- 3.1 Set up development environment
- 3.2 Develop user interface
 - 3.2.1 Implement login and registration
 - 3.2.2 Implement dashboard
 - 3.2.3 Implement navigation components
- 3.3 Develop assessment module
 - 3.3.1 Implement assignment creation
 - 3.3.2 Implement grading functionality
 - 3.3.3 Implement feedback mechanism
- 3.4 Develop communication module
 - 3.4.1 Implement messaging system
 - 3.4.2 Implement discussion forum

3.5 Develop announcement module

3.5.1 Implement announcement creation

3.5.2 Implement announcement display

3.6 Develop submission module

3.6.1 Implement file upload

3.6.2 Implement submission tracking

3.7 Develop learning materials module

3.7.1 Implement material upload

3.7.2 Implement material download

3.7.3 Implement content organization

4.0 Testing

4.1 Develop test plan and test cases

4.2 Perform unit testing

4.3 Perform integration testing

4.4 Perform system testing

4.5 Perform user acceptance testing (UAT)

4.6 Fix defects and retest

5.0 Deployment

5.1 Prepare deployment environment

5.2 Migrate data

5.3 Deploy system to production

5.4 Conduct user training

5.5 Handover system documentation

6.0 Project Management

6.1 Project planning and scheduling

6.2 Monitor and control progress

6.3 Manage budget (RM150,000)

6.4 Risk management

6.5 Communication with stakeholders